SQL

1. **Department Salary Summary**
   select d.dept_id,d.dept_name,count(e.emp_id) as employee_count,coalesce(avg(e.salary),0) as average_salary,coalesce(min(e.salary),0) as minimum_salary,coalesce(max(e.salary),0) as maximum_salary from departments d left join employees e on d.dept_id=e.dept_id group by d.dept_id,d.dept_name;

2. **Second Highest Salary per Department**
   with ranked as (select emp_name,dept_id,salary,dense_rank() over(partition by dept_id order by salary desc) as rnk from employees) select emp_name,dept_id,salary from ranked where rnk=2;

3. **Employees Above Department Average**
   select emp_name,dept_id,salary from employees e where salary>(select avg(salary) from employees where dept_id=e.dept_id);

4. **Manager vs Team Salary**
   select m.emp_name as manager_name,m.salary as manager_salary,count(e.emp_id) as team_count,sum(e.salary) as team_salary from employees m join employees e on m.emp_id=e.manager_id group by m.emp_id,m.emp_name,m.salary having count(e.emp_id)>=2;

5. **Top Project Contributors**
   with r as (select p.project_id,p.project_name,e.emp_id,e.emp_name,ep.hours_worked,dense_rank() over(partition by p.project_id order by ep.hours_worked desc) as rnk from employeeprojects ep join employees e on ep.emp_id=e.emp_id join projects p on ep.project_id=p.project_id) select project_id,project_name,emp_id,emp_name,hours_worked from r where rnk<=2 order by project_id,rnk;

6. **Departments Without Active Projects**
   select distinct d.dept_id,d.dept_name from departments d join employees e on d.dept_id=e.dept_id where not exists (select 1 from projects p where p.dept_id=d.dept_id);

7. **Running Order Revenue**
   with daily_revenue as (select order_date,sum(amount) as daily_revenue from orders where status='completed' group by order_date) select

order_date,daily_revenue,sum(daily_revenue) over(order by order_date) as cumulative_revenue from daily_revenue;

8. **Customer Order Classification**

   with customer_total as (select customer_id,sum(amount) as total_amount from orders where status='completed' group by customer_id) select customer_id,total_amount,case when total_amount>=100000 then 'platinum' when total_amount>=50000 then 'gold' when total_amount>=20000 then 'silver' else 'bronze' end as customer_class from customer_total;

9. **Consecutive Monthly Orders**

10. **Salary Gap Analysis**

    with salary_rank as (select emp_name,dept_id,salary,max(salary) over(partition by dept_id) as highest_salary,min(salary) over(partition by dept_id) as lowest_salary from employees) select emp_name,dept_id,salary,case when salary=highest_salary then 'highest' when salary=lowest_salary then 'lowest' end as salary_type,highest_salary-lowest_salary as salary_difference from salary_rank where salary=highest_salary or salary=lowest_salary;